

KONERU LAKSHMAIAH EDUCATIONAL FOUNDATION

(Deemed to be University)



Course Name: Data Structures and Algorithm 2

Project Title: Power Grid AI

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ABSTRACT

The AI-Based Power Grid Management System is a smart solution designed to improve electricity distribution by combining Artificial Intelligence with Data Structures and Algorithms (DSA). In this project, the power grid is represented as a graph where substations, transformers, and consumers are treated as nodes, while transmission lines are represented as edges. Using graph traversal algorithms such as Dijkstra's shortest path algorithm and Breadth-First Search (BFS), the system identifies the most efficient routes for power transmission, reduces energy loss, and quickly detects faults or damaged lines. Advanced data structures like priority queues and heaps are used to manage power loads, ensuring balanced electricity distribution and minimizing overload conditions.

The AI component of the system analyzes real-time and historical power consumption data to predict electricity demand, detect unusual patterns, and prevent blackouts. Machine learning models help forecast peak usage periods, allowing the grid to distribute energy more efficiently and reduce wastage. Additionally, the system can automatically respond to failures by rerouting electricity through alternative paths, improving reliability and reducing downtime. This project demonstrates how DSA concepts such as graphs, queues, and optimization algorithms can be applied to solve practical problems in modern energy systems while supporting smart, sustainable, and automated power management.